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Investigating the effect of full and partial social media abstinence on fear of missing out and well-being outcomes: A daily diary experimental approach

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Psychology of Popular Media

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Investigating the Effect of Full and Partial Social Media Abstinence on Fear of Missing Out and Well-Being Outcomes: A Daily Diary Experimental Approach

Andree Hartanto[1], K. T. A. Sandeeshwara Kasturiratna[1], Meenakshi Kothari[2], Adalia Y. H. Goh[1], Frosch Y. X. Quek[1], and Nadyanna M. Majeed[3]

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With social media deeply embedded in our daily lives, there is an ongoing debate about its potential negative impact on well-being outcomes. Several lines of correlational research suggest that social media use is associated with reduced well-being. However, these findings are preliminary, heavily relying on cross-sectional data. To address this limitation, our study implemented an experimental paradigm that manipulated varying degrees of social media abstinence and assessed fear of missing out, as well as multiple well-being outcomes, including anxiety, depressive symptoms, life satisfaction, perceived stress, affective states, loneliness, sleep quality, and cognitive failures. A total of 280 regular social media users were randomly assigned to either a full social media abstinence condition, a partial social media abstinence condition, or a control condition for 7 days. During the experimental period, participants were required to complete a daily survey for 7 days to measure their well-being. Participants in the full abstinence condition reported higher levels of fear of missing out. However, our multilevel modeling revealed no significant impact of either full or partial social media abstinence on the well-being outcomes. More importantly, this null effect was consistent across various personality traits, perceived social supports, cyberbullying experience, and individual differences in social media usage, motives, and network. Our findings challenge the widely held assumption that digital detoxes or social media unplugging interventions are effective strategies for enhancing well-being.

Public Policy Relevance Statement

Concerns about the negative impact of social media on well-being have led to widespread recommendations advocating for digital detoxes or unplugging as a method to improve mental health. We conducted an experiment with 280 social media users, having them abstain from social media to varying degrees for 7 days, and measured their fear of missing out and well-being using multiple measures. While participants who fully abstained from social media experienced higher fear of missing out, neither full nor partial social media abstinence improved well-being, regardless of the users' personal characteristics and social media profiles, challenging the assumed effectiveness of digital detox or unplugging interventions.

Keywords: social media abstinence, digital detox, well-being, depression, fear of missing out

Supplemental materials: https://doi.org/10.1037/ppm0000583.supp

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Social media is increasingly ubiquitous in our daily lives—there are over 4.95 billion social media users as of 2023, and this number is expected to rise to 5.42 billion by 2025 (Statista Research Department, 2023). The popularity of social media is not surprising given that it provides an accessible and convenient platform for social connection, information sharing, and creation and transmission of user-generated content (Cheng et al., 2021; Whiting & Williams, 2013). However, with social media being deeply rooted in our daily lives, there is a continuing debate on the possible negative impact of social media on well-being outcomes such as anxiety, depressive symptoms, life satisfaction, self-esteem, and affective states (Hou et al., 2019; Twenge & Campbell, 2019; Vogel et al., 2014; Wong et al., 2024). In particular, social media is argued to have a detrimental psychological impact on users by exposing them to negative stimuli such as unhealthy social comparisons, unpleasant content, cyberbullying, and inducing feelings of inferiority (Gerson et al., 2016; Vogel et al., 2014; Whittaker & Kowalski, 2015). Against this backdrop of possible negative psychological impacts of social media, some have even coined the term 'Facebook depression' to describe the negative psychological effects resulting from social media use (O'Keeffe & Clarke-Pearson, 2011).

Indeed, several lines of cross-sectional research and correlational trend analyses appear to show that social media use is associated with lowered well-being (Boer et al., 2020; Reer et al., 2019; Saiphoo et al., 2020; Valkenburg et al., 2022). These studies have shown an increase in depressive symptoms, perceived stress, and negative affective states in high-intensity social media users (Andreassen et al., 2016; Keles et al., 2020; Lin et al., 2016; Primack et al., 2017; Woods & Scott, 2016). Similarly, other well-being outcomes such as life satisfaction have also been found to decline with higher intensity use of Facebook (Brailovskaia et al., 2023; Chen et al., 2016; Kross et al., 2021). In addition to these empirical studies, several meta-analytic studies have synthesized the existing literature and found a significant negative association between social media and various well-being outcomes (e.g., Appel et al., 2020; Cunningham et al., 2021; Ivie et al., 2020; Marciano et al., 2022; Yoon et al., 2019). For instance, a meta-analysis by Yoon et al. (2019) found that increased time spent on social media was significantly linked to higher levels of depression, albeit with a small effect size. Similarly, a recent meta-analysis by Marciano et al. (2022) reported a positive and small association between social media use and ill-being during the COVID-19 pandemic.

While the negative associations between social media use and well-being appear to be consistent, it is still premature to implicate social media as detrimental to well-being due to the heavy reliance on cross-sectional data in the existing literature (Hartanto et al., 2021; Valkenburg, 2022). This is especially important since the cross-sectional findings are vulnerable to confounds arising from individual difference such as age, socioeconomic status, and personality traits which are commonly associated with both screen time and well-being (Diener et al., 2018; Gambo & Özad, 2020; Gerson et al., 2016; Lönnqvist & große Deters, 2016). Furthermore, the heavy reliance on correlational data severely limits the ability to infer directionality and argues that exposure to social media causes depression (Yoon et al., 2019). Some theories have indeed stressed that the causal relationship might be reversed such that depressive symptoms drive social media use (Hartanto et al., 2021). For example, the theory of compensatory internet use (Kardefelt-Winther, 2014) proposes that people may view online activities such as social media as a means to alleviate negative feelings or fulfill unmet psychosocial needs. Likewise, according to the sociocognitive model of internet addiction (LaRose et al., 2003), given that online and social media stimuli can be psychologically rewarding, people are incentivized to continue engagement, but individuals with poorer self-regulation (e.g., those with psychological disorders) are especially susceptible to developing harmful social media habits. Thus, the use of cross-sectional data in the existing studies oversimplifies our understanding of the etiology and inference surrounding social media and well-being (Keles et al., 2020).

In view of the research gaps, several recent studies have started to employ an experimental paradigm by manipulating social media abstinence to study the effects of social media on well-being outcomes (Brown & Kuss, 2020; Vally & D'Souza, 2019). By experimentally manipulating social media abstinence, the causal effect of smartphone use can be established, and concerns related to reverse causation can be eliminated as random assignment controls for any preexisting differences such as group differences in well-being. Nevertheless, experimental results from these recent social media abstinence studies were mostly mixed. For instance, a few studies have shown that a week-long abstinence from social media led to an increase in life satisfaction and positive affectivity (Brown & Kuss, 2020; Tromholt, 2016), while others have demonstrated that social media abstinence reduced life satisfaction and increased negative affectivity (Vally & D'Souza, 2019; Vanman et al., 2018). Moreover, there were also a few recent studies that found social media abstinence neither increase nor decrease well-being outcomes (Hall et al., 2021; Przybylski et al., 2021; Wilcockson et al., 2019).

There are several factors that may contribute to the mixed findings in the existing social media abstinence studies. First, the mixed findings can be partly driven by the potential differential effect of full and regulated social media use in the form of partial social media abstinence. While some studies relied on full social media abstinence which requires participants to completely abstain from social media (e.g., Hall et al., 2021; Vally & D'Souza, 2019), others have focused on partial abstinence which involves participants to set a time limit for daily social media use (e.g., Hunt et al., 2018). Given that social media can serve as a resource for social support and sense of belonging (Bekalu et al., 2019; Haslam et al., 2017), social media users may experience apprehensions and fear of missing out (FOMO) when abstaining from social media completely (Sekścińska & Jaworska, 2022; Stieger & Lewetz, 2018). This could explain why studies that employed partial social media abstinence are more likely to observe a positive effect of abstinence (Hunt et al., 2018; Stewart, 2020) than those that employed full social media abstinence (Vally & D'Souza, 2019; Vanman et al., 2018). Second, the lack of control in the implementation of social media abstinence may also contribute to the inconsistent findings. Given that social media abstinence can be stressful for some social media users, the reliance on self-report manipulation check in most studies may contribute to lack of compliance (Przybylski et al., 2021; Tromholt, 2016), leading to inconsistent results across studies. Third, a number of social media abstinence studies focused on restricting the use of only one social media platform—such as Facebook (e.g., Turel et al., 2018; Vanman et al., 2018) or Instagram (e.g., Fioravanti et al., 2020). This approach is problematic because it allows participants to access other social platforms such as Twitter and Instagram, which would ultimately weaken

the validity and reliability of the experimental manipulation. Lastly, the mixed findings could also be partly contributed by heterogeneity in social media usage intensity, motives, and network (e.g., number of friends and followers in social media) across the existing studies given that these factors may influence individual's reliance on social media (Andreassen, 2015; Hunt et al., 2021; Schivinski et al., 2020).

Considering these research gaps, the goal of the current study is to build upon previous studies and critically revisit the effects of social media abstinence on FOMO and multiple well-being outcomes, including anxiety, depressive symptoms, life satisfaction, perceived stress, and affective states. We propose two hypotheses: First, given that social media can serve as a resource for social support and a sense of belonging (Bekalu et al., 2019; Haslam et al., 2017), we hypothesize that participants in the full abstinence condition would have higher FOMO compared to those in partial abstinence and control conditions. Second, considering the potential negative psychological impacts of social media, such as exposure to unhealthy social comparisons, unpleasant content, and cyberbullying (Gerson et al., 2016; Vogel et al., 2014; Whittaker & Kowalski, 2015), we hypothesize that both full and partial abstinence would have positive effects on well-being outcomes. However, since FOMO is negatively associated with emotional well-being (Hartanto et al., 2024; Przybylski et al., 2013), we expect the positive effects to be greater in the partial abstinence condition, where lower levels of FOMO are anticipated. We also aim to explore whether the effect of social media abstinence may extend to other important outcomes such as loneliness, sleep quality, and cognitive failures. Four main methodological improvements were incorporated to elucidate the effect of social media abstinence. First, the current study included a novel combination of three conditions: namely, full abstinence condition, partial abstinence condition, and control condition. These conditions enabled us to elucidate the differential psychological effects of full social media abstinence and regulated social media use. Second, we incorporated a more reliable manipulation check for social media abstinence by using an objective screen time tracking system (Hartanto, Chua, et al., 2023; Ohme et al., 2021). Third, to ensure a more comprehensive and valid manipulation of social media abstinence, our manipulation included restrictions on multiple major social media platforms—including Facebook, Twitter, Snapchat, Instagram, Reddit, and Tumblr. Lastly, to examine the boundary conditions of the effects of social media abstinence on FOMO and well-being, we also explored the potential moderating role of various factors, including personality traits, perceived social supports, impression management, cyberbullying experience, and individual differences in social media usage, motives, and network in our analyses.

Method

Participants

A total of 319 individuals participated in the study. Thirty-nine participants either withdrew from the study or were excluded due to failure to adhere to the abstinence instructions, resulting in a final sample of 280 participants (87.77% retention rate, full abstinence = 101, partial abstinence = 87, control = 92). All participants (female = 78%, $M_{age} = 21.78$, $SD = 1.86$) were recruited from a local university in Singapore in exchange for either course credit or S$20 cash. Participants were required to be users of social media for at least 2 hr per day to ensure that only regular users of social media were included in the study. Two hours was selected as the cutoff point as it is considered to be the average time spent by someone who uses social media daily in Singapore (Hartanto, Lee, et al., 2023). All participants provided data for at least 1 day per condition, yielding a total of 1,840 observations out of a possible total of 2,233 (82% completion rate). Data collection was approved by the Institutional Review Board at the authors' university, IRB-21-037-A039-M3(1021). Table S1 in online supplemental materials presents the descriptive statistics for demographics, personality traits, social relationships, social media usage, and need behind social media use of the sample.

Procedure

The study involved a baseline session followed by 7 days of daily diary surveys, employing a between-subject experimental design. Prior to the start of the study, participants were randomly assigned to one of the three conditions (i.e., full abstinence, partial abstinence, or control) and were invited to the respective briefing sessions to ensure that they understood the requirements for participation. During the briefing session, participants in the full abstinence condition were instructed to offload or delete the relevant social media apps. Participants in the partial abstinence condition were instructed to put a screen time limit to 10 min per app per day. Lastly, participants in the control condition were instructed to continue to use social media as usual. The decision to set a fixed limit of 10 min per day, rather than a percentage-based limit, ensured a standardized manipulation within the partial abstinence condition and reduced the likelihood of noncompliance or errors in self-monitoring. After the briefing, a link was emailed to all participants to obtain informed consent. Participants were also required to complete a baseline survey which included questionnaires related to demographics, Big Five personality traits, FOMO, narcissism, perceived social support, cyberbullying victimization, need behind social media use, impression management, and social media popularity. On the following day after the baseline session, the experimental manipulation was carried out for 7 days. During the experimental period, participants were required to complete a daily survey to measure their daily satisfaction, affective state, anxiety, depression, perceived stress, loneliness, self-esteem, and FOMO. The daily survey also included measures of daily stressor exposure which were part of a separate study and thus not analyzed. For each daily survey, participants were allowed to respond within the time window of 8 p.m. that day to 3 a.m. the next day. On the last day (i.e., Day 7), after completing the last daily survey, all participants were required to submit a screenshot of their screen time for the past 7 days.

To measure screen time, iPhone operating system users and Android users used inbuilt applications called Screen Time and Digital Wellbeing, respectively. Both applications are identical in terms of the output of the information in the screenshot and allow participants to set a time limit for the required applications. The screenshots gave an overview of whether the participants adhered to the time limits for both partial abstinence and full abstinence conditions. For example, those in the full abstinence condition sent screenshots of the "most used application" list for the past 7 days, confirming that they fully abstained from the six applications. Participants renamed the screenshots using their unique identification in order to maintain anonymity. Each screenshot was manually checked to ensure its authenticity. After completing the last daily survey, participants were compensated and completed an honesty check survey to ensure the validity of their responses. Participants

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were asked if they provided honest responses throughout the course of the study and whether they complied with the experimental instructions. They were informed that their responses were anonymous and that they would still receive compensation even if they indicated they were not entirely honest. Only participants who passed the honesty check were included in the final analysis.

Materials

Baseline

Trait FOMO. Trait FOMO was assessed using the 10-item Fear of Missing Out Scale (Przybylski et al., 2013). Participants indicated their level of agreement with various statements about social situations (e.g., "I fear others have more rewarding experiences than me") on a 5-point scale (1 = *not at all true of me* to 5 = *extremely true of me*; $\alpha = .90$).

Trait Narcissism. Trait narcissism was assessed using the seven-item Narcissism subscale of the Short Dark Tetrad Scale (Paulhus et al., 2021). Participants were asked to rate their agreement with each statement (e.g., "People see me as a natural leader") on a 5-point scale (1 = *strongly disagree* to 5 = *strongly agree*; $\alpha = .74$).

Big Five Personality Traits. Big Five personality traits were assessed using the 20-item Short International Personality Item Pool-Big Five Markers-20 Questionnaire (Topolewska et al., 2014). Participants rated their agreement with four items each across five personality dimensions, namely, Agreeableness ("Make people feel at ease"; $\alpha = .72$), Emotional Stability ("Get irritated easily"; $\alpha = .69$), Extraversion ("Don't talk a lot"; $\alpha = .77$), Conscientiousness ("Follow a schedule"; $\alpha = .76$), and Openness to Experience ("Have excellent ideas"; $\alpha = .65$), on a 5-point scale (1 = *very inaccurate* to 5 = *very accurate*).

Impression Management Trait. Impression management was assessed using the 13-item Marlowe–Crowne Social Desirability Scale (Reynolds, 1982). Participants evaluated each statement (e.g., "Sometimes I try to get even, instead of forgive and forget") by rating them as either true or false in relation to their behavior ($\alpha = .64$).

Social Support. Perceived social support was assessed using the 12-item Multidimensional Scale of Perceived Social Support (Zimet et al., 1988). Participants were asked to rate the quality of social support they received with respect to a significant other (e.g., "There's a special person around when I am in need"; $\alpha = .95$), friends (e.g., "I can count on my friends when things go wrong"; $\alpha = .91$), and family (e.g., "My family really tries to help me"; $\alpha = .91$) on a 7-point scale (1 = *very strongly disagree* to 7 = *very strongly agree*).

Needs Behind Social Media Use. The motivations behind using social media were measured using the 20-item Social Networking Sites Usage and Needs Scale (Ali et al., 2020). The scale covers five domains, including diversion needs (e.g., "Social media helps me to feel less lonely"; $\alpha = .91$), cognitive needs (e.g., "Social media helps me in research and studies"; $\alpha = .72$), affective needs (e.g., "Using social media is one of the routine things I do when I am online"; $\alpha = .67$), personal integrative needs (e.g., "Social media is a part of my self-image"; $\alpha = .82$), and social integrative needs (e.g., "Social media allows me to communicate with my friends"; $\alpha = .70$). Participants were asked to rate their agreement with each statement on a 5-point scale (1 = *strongly disagree* to 5 = *strongly agree*).

Social Media Usage. Social media usage was assessed using two distinct indicators. The first indicator involved a question about the daily time spent on social media (i.e., "How much time (in hours) do you spend on social media per day?"). The second indicator focused on the daily frequency of social media checking (i.e., "How many times do you check any social media applications per day?"). Participants provided their answers to these questions in a numerical format, allowing for a free response to accurately reflect their social media usage.

Social Media Network Size. Social media network size was assessed through a series of questions to estimate the participants' social media network on four prominent social media platforms, including Facebook ("How many Facebook friends do you have?"), Instagram ("How many Instagram followers do you have?" and "How many people do you follow on Instagram?"), Twitter ("How many people do you follow on Twitter?" and "How many Twitter followers do you have?), and Snapchat ("How many streaks do you have on Snapchat?"). Given that participants may use multiple social media platforms and have different profiles of social media usage, the responses across these platforms were aggregated to create an index reflecting the breadth of each participant's social media network.

Cyberbullying Victimization. Cyberbullying victimization was assessed using the 27-item cyberbullying victimization scale (Lee et al., 2017). Participants were asked to rate on statements related to the frequency of cyberbullying experiences on a 5-point scale (1 = *not at all* to 5 = *very often*; $\alpha = .90$).

Daily

Daily State FOMO. Daily state FOMO was measured through a single-item question, "Today, I experienced fear of missing out." Participants rated their level of agreement with this statement on a 5-point Likert scale ranging from 1 = *strongly disagree* to 5 = *strongly agree*.

Daily Life Satisfaction. Daily life satisfaction was assessed using the five-item Quality of Day Measure (Diener et al., 1985). Participants were asked to rate their agreement with each statement (e.g., "In most ways, today was close to ideal"; "Today was an awful day") on a 7-point Likert scale (1 = *strongly disagree* to 7 = *strongly agree*; control $\alpha_{7\ \text{days}} = [.84, .90]$, partial abstinence $\alpha_{7\ \text{days}} = [.84, .89]$, and full abstinence $\alpha_{7\ \text{days}} = [.82, .89]$).

Daily Affective State. Daily affective state was assessed using the 10-item International Positive and Negative Affect Schedule Short Form (Thompson, 2007). The scale includes five items to measure positive affect (e.g., "active" and "attentive"; control $\alpha_{7\ \text{days}} = [.80, .88]$, partial abstinence $\alpha_{7\ \text{days}} = [.81, .87]$, and full abstinence $\alpha_{7\ \text{days}} = [.81, .87]$) and five items for negative affect (such as "hostile" and "ashamed"; control $\alpha_{7\ \text{days}} = [.76, .89]$, partial abstinence $\alpha_{7\ \text{days}} = [.76, .89]$, and full abstinence $\alpha_{7\ \text{days}} = [.73, .83]$). Participants rated these items on a 5-point scale (1 = *never* to 5 = *always*) to reflect their affective state on the present day.

Daily State Anxiety. Daily state anxiety was assessed using the six-item Shortened Spielberg State-Trait Anxiety Inventory (Marteau & Bekker, 1992). Participants evaluated their level of anxiety on the present day by responding to anxiety-related statement (e.g., "I feel calm," "I feel tense") on a 4-point scale (1 = *not at all* to 4 = *very much*; control $\alpha_{7\ \text{days}} = [.80, .88]$, partial abstinence $\alpha_{7\ \text{days}} = [.81, .87]$, and full abstinence $\alpha_{7\ \text{days}} = [.80, .87]$).

Daily Depressive Symptoms. Daily depressive symptoms were assessed using the 10-item Centre for Epidemiologic Studies

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Depression Scale-Revised (Andresen et al., 1994). Participants were asked to rate their frequency of experiencing the depressive symptoms described in each statement on the present day (e.g., "I was bothered by things that usually don't bother me," "I felt depressed") on a 4-point scale (0 = *rarely or none of the time* to 3 = *all the time*; control $\alpha_{7\ \text{days}} = [.80, .86]$, partial abstinence $\alpha_{7\ \text{days}} = [.80, .87]$, and full abstinence $\alpha_{7\ \text{days}} = [.78, .85]$).

Daily Perceived Stress. Daily perceived stress was assessed using the four-item Shortened Perceived Stress Scale (Cohen et al., 1983). Participants were asked to rate their frequency of experiencing the stress-related scenarios described in each statement on the present day (e.g., "How often have you felt you were unable to control the important things in life?") on a 5-point scale (0 = *never* to 4 = *very often*; control $\alpha_{7\ \text{days}} = [.61, .74]$, partial abstinence $\alpha_{7\ \text{days}} = [.67, .78]$, and full abstinence $\alpha_{7\ \text{days}} = [.61, .78]$).

Analytic Plan

The current study used multilevel modeling to examine the effect of social media abstinence (i.e., partial abstinence vs. control and full abstinence vs. control) on FOMO, emotional well-being, and other outcomes. As the study consisted of repeated measures over a period of 7 days, multilevel modeling is a more apt method of analysis than ordinary least squares regression as it helps to take into account the nonindependence of data within participants while considering multiple data points at different levels (Field & Wright, 2011). This is also in conjunction with Heck and Thomas (2020), who emphasized the advantages of using multilevel modeling to overcome the limitations of other methods such as traditional linear models (e.g., repeated measures analysis of variance [ANOVA]) as it is able to disentangle within-person and between-person heterogeneity by allowing for the addition of a random component at each level. The equation used for the analysis is as follows:

$$\text{Level 1: } (\text{Outcome})_{di} = B_{0i} + \varepsilon_{di};$$

$$\text{Level 2: } B_{0i} = \gamma_{00} + \gamma_{01}(\text{full abstinence})_i + \gamma_{02}(\text{partial abstinence})_i + \mu_{0i}.$$

In the above equation, the fixed parameter γ_{00} represents the intercept value indicating the average level of the outcome variable across all participants for the control condition, while the random parameter μ_{0i} represents the participant-level deviation from the aforementioned sample-level intercept for each participant *i*. Lastly, ε_{di} represents the residual (i.e., the difference between the daily predicted and observed levels of the outcome in question). Additionally, the current study used mixed ANOVA to examine the effect of condition on FOMO, emotional well-being, and other outcome variables measured on each day of the study, in order to analyze the pattern of changes that take place within the 7-day period for each variable.

Transparency and Openness

The current study's design and its analysis plan were not preregistered. The relevant materials, data set, and R analytic code, as well as full summaries of the results, have been made publicly available on ResearchBox (744; https://researchbox.org/744). All analyses were conducted in R Version 3.6.3 (R Core Team, 2020). Scale reliabilities were calculated using psych Version 2.2.5 (Revelle, 2022). Effect sizes were calculated in the form of standardized coefficients for fixed effects using effectsize Version 0.6.0.1 (Ben-Shachar et al., 2020). Multilevel modeling was conducted using lme4 Version 1.1-28 (Bates et al., 2015), and significance testing was carried out via lmerTest Version 3.1-3 (Kuznetsova et al., 2017).

Results

State FOMO and Daily Well-Being Outcomes

A series of multilevel models were estimated to examine the effect of social media abstinence (i.e., partial abstinence vs. control and full abstinence vs. control) on daily FOMO and well-being indicators: life satisfaction, positive affect, negative affect, state anxiety, depressive symptoms, loneliness, and perceived stress (see Table 1). Consistent with our hypothesis, we found a significant effect of full abstinence on FOMO as compared to control ($\gamma_{10} = -.29$, $SE = 0.12$, $\beta = -.15$, 95% confidence interval [CI] = [−.28, −.03], $p = .017$). Individuals in the full abstinence condition ($M = 2.27$, $SD = 1.20$) reported higher levels of daily FOMO as compared to individuals in the control condition ($M = 2.00$, $SD = 1.10$). However, there was no significant effect of partial abstinence ($M = 2.08$, $SD = 1.22$) on FOMO as compared to control ($\gamma_{10} = -.18$, $SE = 0.12$, $\beta = -.09$, 95% CI = [−.22, .03], $p = .145$).

For our daily well-being outcomes, multilevel modeling indicated that, as compared to the control condition, there was no significant effect of fully abstaining from social media abstinence on daily life satisfaction ($\gamma_{10} = .11$, $SE = 1.33$, $\beta = .05$, 95% CI [−0.07, 0.16], $p = .420$), positive affect ($\gamma_{10} = -.13$, $SE = 0.10$, $\beta = -.11$, 95% CI [−0.27, 0.05], $p = .190$), negative affect ($\gamma_{10} = -.08$, $SE = 0.07$, $\beta = -.09$, 95% CI [−0.24, 0.06], $p = .240$), state anxiety ($\gamma_{10} = -.08$, $SE = 0.06$, $\beta = -.07$, 95% CI [−0.20, 0.05], $p = .240$), depressive symptoms ($\gamma_{10} = -.10$, $SE = 0.05$, $\beta = -.13$, 95% CI [−0.28, 0.02], $p = .081$), and perceived stress ($\gamma_{10} = -.03$, $SE = 0.08$, $\beta = -.03$, 95% CI [−0.17, 0.11], $p = .700$). Similarly, we did not find any significant effect of partially abstaining from social media, as compared to the control condition, on all well-being outcomes: daily life satisfaction ($\gamma_{10} = -.20$, $SE = 0.14$, $\beta = -.08$, 95% CI [−0.20, 0.03], $p = .160$), positive affect ($\gamma_{10} = -.01$, $SE = 0.10$, $\beta = -.01$, 95% CI [−0.18, 0.15], $p = .890$), negative affect ($\gamma_{10} = -.06$, $SE = 0.07$, $\beta = -.06$, 95% CI [−0.21, 0.09], $p = .420$), state anxiety ($\gamma_{10} = -.01$, $SE = 0.07$, $\beta = -.01$, 95% CI [−0.13, 0.12], $p = .920$), depressive symptoms ($\gamma_{10} = -.06$, $SE = 0.06$, $\beta = -.08$, 95% CI [−0.23, 0.07], $p = .289$), and perceived stress ($\gamma_{10} = .05$, $SE = 0.08$, $\beta = .05$, 95% CI [−0.09, 0.19], $p = .510$). In addition, a series of repeated measures ANOVAs were performed to examine the effect of full and partial social media abstinence for each day. We did not find any significant difference in all well-being indicators between participants in full abstinence condition, partial abstinence, and control on any of the 7 days of the study ($ps < .05$).

Exploratory Moderation Analysis

We conducted exploratory moderation analyses to test the moderating role of trait FOMO, trait narcissism, the big five personality traits, quality of relationships, need behind social media use, impression management, and cyberbullying victimization on the effect of full and partial abstinence from social media on FOMO and well-being outcomes. Overall, none of the moderators have a consistent

Table 1
Summary of Fixed Effects of Social Media Abstinence on FOMO and Well-Being Outcomes

	Full (vs. control)					Partial (vs. control)				
Outcome	β	95% CI	γ_{10}	*SE*	*p*	β	95% CI	γ_{10}	*SE*	*p*
FOMO	−.15	[−0.28, −0.03]	−.29	0.12	.017	−.09	[−0.22, 0.03]	−.18	0.12	.145
Life satisfaction	.05	[−0.07, 0.16]	.11	1.33	.420	−.08	[−0.20, 0.03]	−.20	0.14	.160
Positive affect	−.11	[−0.27, 0.05]	−.13	0.10	.190	−.01	[−0.18, 0.15]	−.01	0.10	.890
Negative affect	−.09	[−0.24, 0.06]	−.08	0.07	.240	−.06	[−0.21, 0.09]	−.06	0.07	.420
State anxiety	−.07	[−0.20, 0.05]	−.08	0.06	.240	−.01	[−0.13, 0.12]	−.01	0.07	.920
Depression	−.13	[−0.28, 0.02]	−.10	0.05	.081	−.08	[−0.23, 0.07]	−.06	0.06	.289
Perceived stress	−.03	[−0.17, 0.11]	−.03	0.08	.700	.05	[−0.09, 0.19]	.05	0.08	.510

Note. FOMO = fear of missing out; β = standardized slope coefficient; 95% CI = 95% confidence interval of β; γ_{10} = unstandardized slope coefficient of gratitude condition; *SE* = standard error of γ_{10}.

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moderating effect on FOMO and well-being outcomes. See Tables S4, S5, and S6 in online supplemental materials for the full results.

Exploratory Outcomes

We also conducted exploratory analyses on three potential outcomes, including daily loneliness, cognitive failure, and sleep quality (see the online supplemental materials for details on the exploratory measures). Our multilevel modeling showed that, as compared to the control condition, there was no significant effect of fully abstaining from social media on daily loneliness ($\gamma_{10} = -.09$, $SE = 0.08$, $\beta = -.08$, 95% CI [−0.21, 0.06], $p = .270$), cognitive failures ($\gamma_{10} = -.14$, $SE = 0.07$, $\beta = -.19$, 95% CI [−0.37, 0.00], $p = .053$), and perceived sleep quality ($\gamma_{10} = -.02$, $SE = 0.02$, $\beta = -.06$, 95% CI [−0.16, 0.05], $p = .300$). Similarly, as compared to the control condition, we also did not find evidence that partially abstaining from social media has significant effect on daily loneliness ($\gamma_{10} = -.09$, $SE = 0.09$, $\beta = -.07$, 95% CI [−0.21, 0.06], $p = .290$), cognitive failures ($\gamma_{10} = -.11$, $SE = 0.07$, $\beta = -.14$, 95% CI [−0.33, 0.05], $p = .138$), and perceived sleep quality ($\gamma_{10} = -.02$, $SE = 0.02$, $\beta = -.04$, 95% CI [−0.15, 0.06], $p = .440$).

Discussion

Social media has become a popular platform for social connection and information sharing. However, given that social media may expose users to cyberbullying, negative contents, and feelings of inferiority, along with its increasing popularity, this has led to growing concerns about the potential negative consequences of social media on well-being outcomes (Gerson et al., 2016; Vogel et al., 2014; Whittaker & Kowalski, 2015). While a number of studies have shown negative correlations between social media use and well-being (Hou et al., 2019; Twenge & Campbell, 2019; Vogel et al., 2014), the heavy reliance on correlational data severely limits the ability to infer the causal effect of social media. Thus, the current study aimed to critically examine the negative implications of social media by experimentally testing the effect of full and partial social media abstinence on multiple well-being indicators. Building upon previous studies, our study differentiated between full and partial social media abstinence, incorporated a reliable manipulation check using an objective screen time tracking system, restricted usage across multiple major social media platforms, and explored the moderating roles of various factors, including personality traits, perceived social supports, impression management, cyberbullying experience, and individual differences in social media usage, motives, and network.

Contrary to our hypothesis, we did not find any evidence that either full or partial social media abstinence had any significant effect on participants' well-being. Specifically, we found that the anxiety, depressive symptoms, life satisfaction, self-esteem, and affective states of those who abstained from social media were similar to those who were allowed to use each social media for 1 hr per day and those who were allowed to use social media freely. The results are consistent across 7 days of social media abstinence, suggesting that the number of days of abstinence did not have a cumulative influence on all of the well-being indicators. Beyond the lack of main effect of full and partial social media abstinence on well-being, we also found that the hours of social media usage, frequency of social media login, number of social networks in social media, and motivations behind social media use did not significantly moderate any effect of social media abstinence. The results cast doubt on the possibility that the mixed findings in the existing social media abstinence studies (Brown & Kuss, 2020; Hall et al., 2021; Tromholt, 2016; Vally & D'Souza, 2019; Wilcockson et al., 2019) were driven by heterogeneity in personality traits, social relationships, cyberbullying experience, and social media usage intensity, motives and network (e.g., number of friends and followers in social media) across the existing studies.

While we did not find any effect of social media abstinence on well-being indicators, full social media abstinence was found to elevate FOMO in our participants. The heightened FOMO suggests that the inability to use social media for 7 days has created perceptions of missed opportunities to connect with others through social media (Roberts & David, 2020; Uram & Skalski, 2022). The findings support the critical role of social media in daily interactions and the constant connectivity it offers. Given the negative link between FOMO and well-being (Hartanto & Kasturiratna, 2024; Przybylski et al., 2013), it is also plausible that the heightened FOMO in the full social media abstinence condition may undermined any potential benefits of full social media abstinence. This alternative explanation may also account for the null effects observed in our well-being measures.

Our null experimental findings also raised doubt about the directionality of the previous cross-sectional studies that showed a negative correlation between social media use and well-being (Boer et al., 2020; Reer et al., 2019; Saiphoo et al., 2020). The null effect found in the current experimental study suggests that lowered well-being—commonly indicated by depressive symptoms, negative affect,

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anxiety, and lack of positive affect and self-esteem—from the previous cross-sectional studies were likely the antecedent rather than consequences of social media usage (Hartanto et al., 2021). This reverse causation perspective is in line with several previous studies that similarly failed to find an effect on well-being indicators such as life satisfaction and depressive symptoms (Hall et al., 2021; Przybylski et al., 2021; Wilcockson et al., 2019). The reverse causation perspective is also consistent with the theory of compensatory internet use, which proposes that people may view online activities such as social media as a means to alleviate negative feelings or fulfill unmet psychosocial needs (Kardefelt-Winther, 2014). Our findings may also align with the HomeNet studies of the late 1990s, which initially found negative effects of increased internet use on well-being (Kraut et al., 1998) but later showed these effects dissipated over time influenced by personal characteristics, user experience, and the type of internet use (Kraut et al., 2002). Similar to the HomeNet studies, it is possible that any initial negative effects of social media use may dissipate over time as users become more experienced with social media.

There are several noteworthy directions for future research. First, while our sample consisted solely of young adults, future studies should include a broader age range to enhance the generalizability of findings. While numerous nationwide surveys have shown that young adults use the most social media (Auxier & Anderson, 2021; Gambo & Özad, 2020; Murthy et al., 2016), there is an increasing need to understand the psychological consequences of social media in children and older adults given the growing trend of social media usage in these two age groups (Anderson & Perrin, 2017; Hunsaker & Hargittai, 2018). Second, the duration of our experimental manipulation was limited to 1 week. Future studies could explore longer periods of social media abstinence to determine if more extended durations reveal more pronounced psychological effects, either positive or negative, which a shorter duration might not capture. Additionally, the fixed time limit of 10 min per day for the partial abstinence condition may have introduced variability in how restrictive the condition was for different participants. Heavier users of social media might have found the 10-min limit more restrictive compared to lighter users, potentially introducing noise within the condition. Future research could address this limitation by implementing a percentage-based limit to standardize the reduction relative to each participant's typical usage.

Nevertheless, several strengths of the current study are noteworthy as well. For instance, the incorporation of a reliable manipulation check with an objective screen time tracking system and honesty check allowed us to rule out the possibility of noncompliance in our experimental manipulation. Furthermore, the restriction of multiple major social media platforms reduced the possibility that participants were able to access other social platforms. The careful and controlled implementation of our manipulation could explain why we found a robust effect of social media abstinence on FOMO despite the null effect in all of our indicators of well-being. Furthermore, numerous studies have also revealed that what people say about their media usage often does not align with their actual usage patterns (Greenberg et al., 2005; Kahn et al., 2014; LaRose & Eastin, 2004; Parry et al., 2021) and the discrepancy between intended or self-reported media use and actual or objective media usage can significantly impact the research outcomes (e.g., Burke & Kraut, 2016; Sewall et al., 2022). Our experiment addresses this issue not only by focusing on social media abstinence as an actual behavior but also by utilizing objective measures of social media use to ensure compliance. This methodological strength further supports the robustness of our findings and highlights the value of using objective data and real behavioral interventions to gain a comprehensive understanding of media use and its impacts. Lastly, the high sample size in the current study increases the statistical power to detect small to medium effect sizes.

In view of the concerns about the possible negative impact of social media on well-being, numerous sources in the mass media, including newspaper articles, websites, blogs, and self-help books, have proposed digital detox or unplugging as a trendy and effective intervention to improve overall well-being (Jorge, 2019; Price, 2018; Radtke et al., 2022; Shlain, 2019). Although it is widely assumed that a short digital detox can improve well-being, our study found no evidence supporting the effects of either full or partial social media abstinence on well-being outcomes. These results have important practical implications, challenging the assumed effectiveness of digital detox or unplugging interventions widely practiced by social media users.

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